Nathan Sankaran

DATA SCIENTIST | MACHINE LEARNING ENGINEER | AI ENGINEER
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EDUCATION

University at Buffalo, The State University of New York **Aug 2025 - Dec 2026**
Master of Science, Engineering Science (Data Science) (GPA: 3.82) *Buffalo, NY*

Vellore Institute of Technology **Aug 2019 - Jul 2023**
Bachelor of Technology, Computer Science and Engineering with Specialization In AI and ML *Chennai, IN*

WORK EXPERIENCE

Zolvit (formerly Vakilsearch) | *Machine Learning Engineer Intern* **Feb 2024 - Aug 2024**

- Eliminated ~23 hours of weekly manual data entry by engineering a fully automated ingestion pipeline for 350+ complex documents, combining Google Vision OCR with a fine-tuned T5-large model to replace a fragile legacy process.
- Drove 99.9% service uptime and eliminated recurring outages by deploying Dockerized ML microservices on AWS EC2 with FastAPI, restart policies, and real-time monitoring via Grafana.
- Automated routing of customer-uploaded documents across categories by achieving 92% classification accuracy, benchmarking Logistic Regression, SVM, and XGBoost on TF-IDF and Doc2Vec features extracted via Google Vision OCR.
- Improved legal precedent retrieval precision by ~28% for lawyers by building a hybrid search engine combining Elasticsearch keyword search with Pinecone vector embeddings across 10,000+ documents.

PROJECTS

SEC Earnings Extraction: LLM Benchmark, Fine-Tuning and Serving | GitHub **Oct 2026 - Nov 2026**

- Built a pipeline linking 4,700 SEC earnings releases (non-financial S&P 500, 2023–2026) to the XBRL figures in each company's later 10-Q/10-K, plus an auto-grader that scores LLM-extracted revenue, net income and EPS and classifies every error.
- Benchmarked 9 LLMs (5 API, 4 open-weight on vLLM); found 80% of open-model errors were unit slips, and a reply format of figure + unit, converted in code, cut open-model errors 65% (Llama-3.1-8B: 44.4% → 85.0%).
- Fine-tuned Qwen3-8B with LoRA on 2,000 releases labelled automatically from XBRL; on 37 companies unseen in training, accuracy rose from 89.1% to 99.2% (+10.1 pts, 95% CI +5.1 to +15.6), with the test protocol fixed before any test-set prediction.
- Load-tested 4 vLLM serving variants on one H100 (LoRA, merged, FP8, 4-bit AWQ): FP8 served 37% more releases per hour (6,143; $0.56 per 1,000 vs $1.24–$24.11 for APIs) at near-identical accuracy; 4-bit cut weight memory 2.6x without a throughput gain.

US Electricity Demand Forecast Audit | GitHub **Oct 2026 - Nov 2026**

- Audited 10 years of day-ahead electricity demand forecasts for 50+ US grid operators, analyzing 6.1M hourly time-series records of public U.S. Energy Information Administration (EIA) data in Python and pandas.
- Found 3 errors in EIA's published data, verified against grid operators' own records, that made forecast accuracy appear to decline; once corrected, the 10-year trend was flat.
- Traced California's apparent forecast failure to a 1-hour timestamp shift and battery charging counted as demand (R-squared 0.91); corrected, its 2025 error fell from 8.3% to 2.4%, beating a naive baseline.
- Pre-registered the analysis plan before computing results and built a reproducible pipeline with 138 automated tests (pytest), data-quality rules, statistical trend tests and 8 robustness checks.

Compositional Grounding Probes for Open-Vocabulary Segmentation | GitHub **Aug 2026**

- Built a 914-item benchmark from COCO, LVIS, and Visual Genome to evaluate text-prompted segmentation, with automatically generated labels individually verified by a human reviewer.
- Exposed systematic model failures on compositional prompts: 20.7% accuracy on negation vs 93% on controls, and 0 of 51 correct on paired spatial queries.
- Traced all failures to the text-grounding stage via an oracle ablation that scored 99.6% with ground-truth inputs.

Drift-Aware Retraining: Experiment and MLOps Pipeline | GitHub **Feb 2026**

- Replayed $4.2B of Lending Club loans (36 months, 8 retraining policies, 1–6 month label delays): at a 10% decline budget, retraining declined $8–14M more in loans that later defaulted; quarterly matched monthly's gain within 2% with 69% fewer retrains.
- Found drift triggers (KS, domain classifier) needed ~60% more retrains than quarterly for the same result and missed injected concept drift; a label-based calibration test caught 100% at 89–97% alarm precision. Built the FastAPI, Airflow, MLflow pipeline.

SKILLS

- **Programming Languages**: Python, C/C++, SQL, R
- **Machine Learning & AI**: PyTorch (torchvision, PyTorch Geometric), TensorFlow, Scikit-learn, XGBoost, Hugging Face, embeddings, RAG, Representation Learning (self-supervised & contrastive learning), LangChain, LangGraph, TensorBoard, PEFT (LoRA), vLLM
- **Data Processing & Analysis**: Pandas, NumPy, Medical Imaging (CT, DICOM, pydicom, volumetric data), RDKit
- **Data Visualization & BI**: Matplotlib, Seaborn, Plotly, Streamlit, Tableau, PowerBI
- **MLOps & Cloud**: Docker, Kubernetes, AWS, Apache Airflow, MLflow, Evidently AI, Prometheus, Grafana
- **Developer Tools & Systems**: Linux, Git/GitHub, Pytest, REST APIs, CUDA/GPU training, Coding Agents (Cursor, Claude Code, Codex)
- **Databases & Data Engineering**: PostgreSQL, MySQL, Elasticsearch, Pinecone, Spark, Snowflake

AWARDS

- Felix Infausto Scholarship - University at Buffalo, 2025